

Packets Primer

 **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

Download the packet capture file and use packet analysis software to find the flag.

Hints

- Wireshark, if you can install and use it, is probably the most beginner friendly packet analysis software product.
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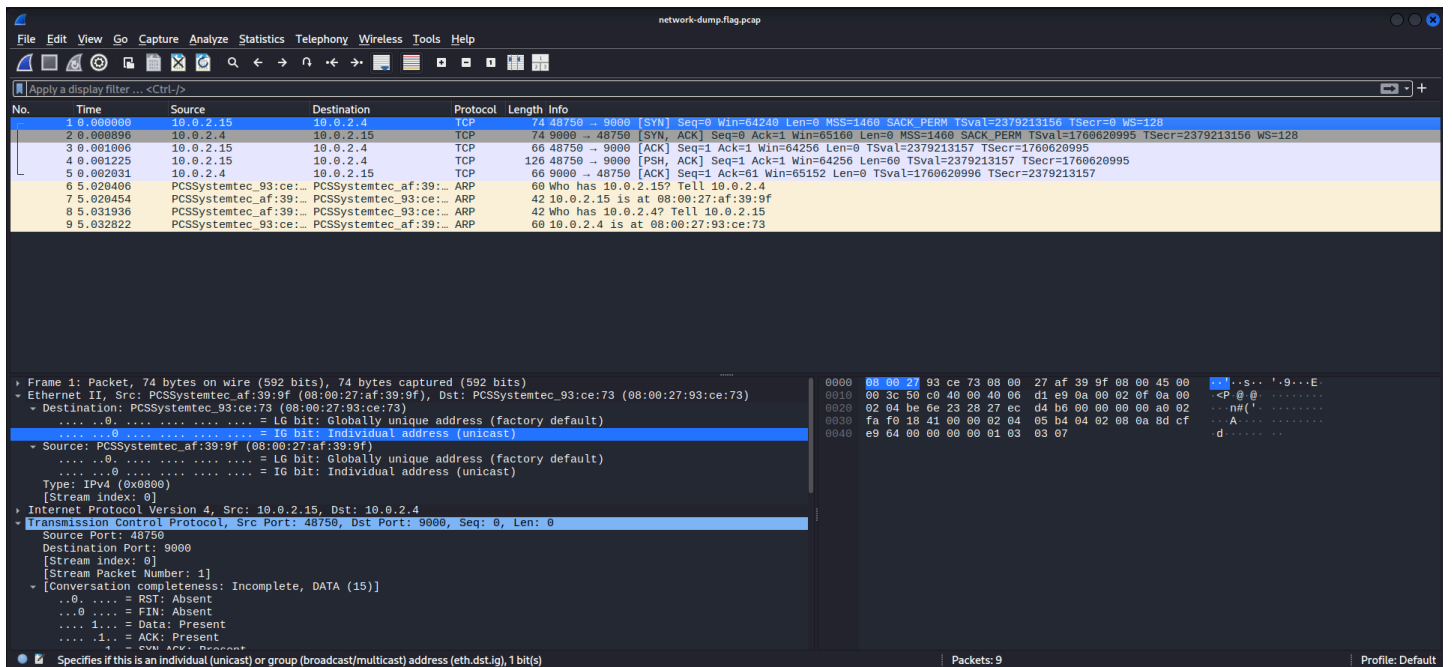
Tools Used

- `file` – Determine file type
 - `strings` – Extract printable character sequences from binary files
 - `grep` – Filter output for specific patterns
 - **Wireshark** – GUI tool for analyzing network traffic
 - **Browserling** – Online tool to remove whitespace from text
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Complete Solution

Step 1: Download and Prepare

Download the PCAP file named `network-dump.flag.pcap` from the challenge page.



Step 2: Basic File Inspection

First, verify the file type and general information:

```
file network-dump.flag.pcap
```

Expected output:

```
network-dump.flag.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4 (Ethernet, capture length 262144)
```

The file is a network packet capture (PCAP) containing Ethernet traffic.

Step 3: Quick String Search

Sometimes flags are directly visible as plain text strings within the binary data. Let’s try a simple `strings` scan:

```
strings network-dump.flag.pcap | grep -i "picoCTF"
```

Output: (no results)

The flag doesn't appear as a continuous string. However, upon closer inspection of the packet capture, we notice something interesting...

Step 4: Manual Packet Inspection

When examining the individual packets in Wireshark, we observe the flag hidden across multiple packets, with spaces between each character:

```
p i c o C T F { <REDACTED> }
```

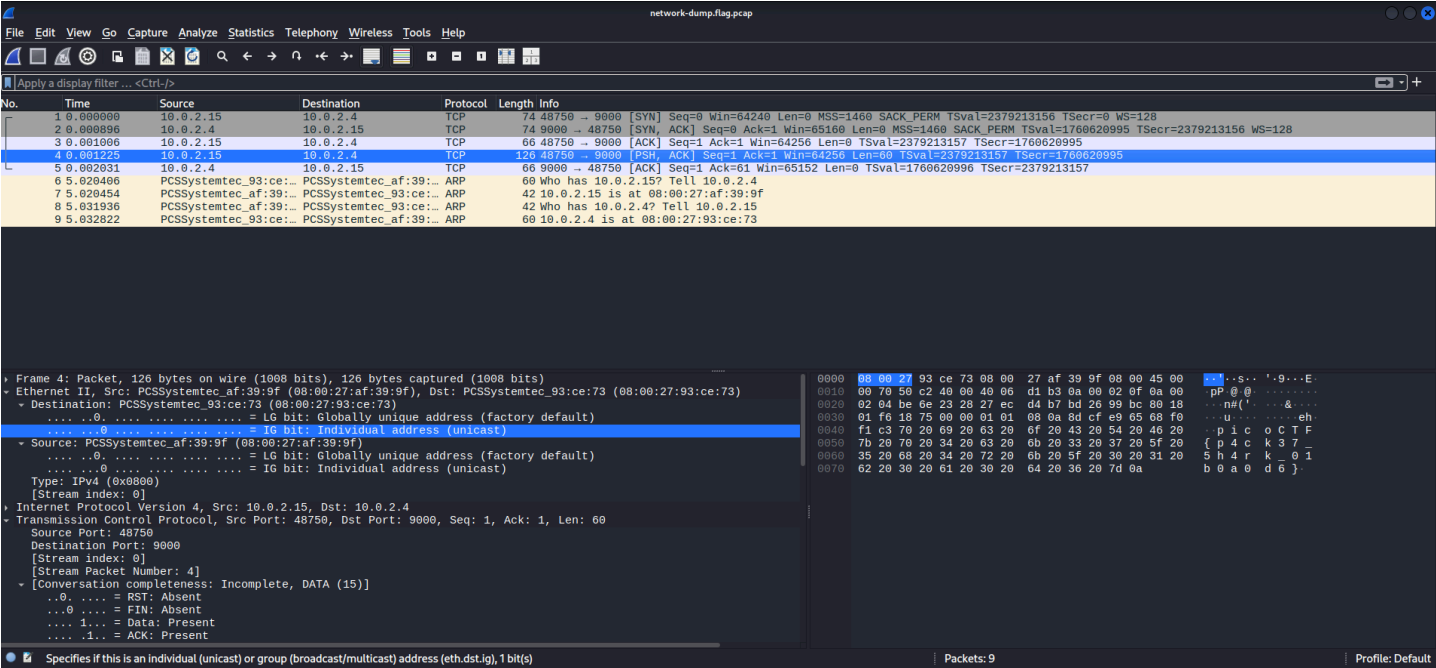
The flag characters are scattered throughout the capture, separated by spaces.

Step 5: Reconstruct the Flag

To get the actual flag, we need to **remove all the spaces** from the scattered text.

You can do this using an online tool like [Browserling's Remove All Whitespace Tool](#):

- 1. Copy all the visible text from the packet capture (including the scattered flag characters)
- 2. Paste it into the online tool
- 3. Click "Remove All Whitespace"



Result:

picoCTF{<REDACTED>}

Step 6: Alternative – Remove Spaces with Python

If you prefer a command-line solution, you can also use a simple Python one-liner:

```
echo "p i c o C T F { <REDACTED> }" | python3 -c "import sys;
print(''.join(sys.stdin.read().split()))"
```

Or save the scattered text to a file and process it:

```
# remove_spaces.py
with open('scattered_flag.txt', 'r') as f:
    text = f.read()
    flag = ''.join(text.split())
    print(flag)
```

python



Final Result

picoCTF{<REDACTED>}

Note: The actual flag is redacted here. In the real challenge, removing the spaces from the scattered text will reveal the complete flag.



Key Concepts

Concept	Description
PCAP Analysis	Network packet captures contain raw packet data that may contain flags or other sensitive information.

Concept	Description
Strings Extraction	A powerful technique for finding readable text embedded in binary files.
Scattered Data	Flags can be split across multiple packets or fields, requiring reconstruction.
Whitespace Removal	When characters are separated by spaces, removing the spaces reconstructs the original string.
Manual Inspection	Sometimes the most effective approach is simply reading the packets in Wireshark.

Pro Tips

1. **Always start with Wireshark** – A visual inspection can reveal patterns that automated tools might miss.
2. **Look for scattered characters** – Sometimes flags are intentionally broken up to avoid detection by simple `strings` searches.
3. **Use online tools for quick fixes** – Removing whitespace is a common operation; online tools can save time.
4. **Don't overcomplicate** – Before writing complex scripts, try the simplest solution (copy-paste into a text editor and use find/replace to remove spaces).
5. **Check every field** – Flags can appear in packet payloads, but also in protocol headers, comments, or other metadata fields.